POWERFLEX 525 AC DRIVE ? USER MANUAL

Firmware Version 11.02
Catalog Numbers: 25B-D1P4N104 through 25B-D480N104

1. INTRODUCTION

The PowerFlex 525 is a compact, multi-drive AC variable frequency drive designed for applications from 0.4 kW to 75 kW. It features built-in EtherNet/IP, dual-port Ethernet, and a removable HIM (Human Interface Module). This manual covers installation, configuration, parameter programming, troubleshooting, and maintenance.

1.1 SAFETY

WARNING: The drive contains hazardous voltages. The DC bus capacitors retain dangerous voltage for up to 5 minutes after power is removed. Always verify that the DC bus voltage is below 50V DC before touching any internal components.

1.2 PARAMETER GROUP OVERVIEW

Parameters are organized into groups identified by a letter prefix:

- b ? Basic Display: read-only status and monitoring parameters
- A ? Advanced Program: configuration and tuning parameters
- C ? Communications: network and serial port settings
- d ? Data: diagnostic and data logging parameters
- F ? Fault and Diagnostic: fault history and diagnostic data
- G ? Blower/Fan: cooling fan control parameters
- P ? Program: basic motor and drive program parameters
- t ? Timer/Counter: application timer and counter parameters

1.3 BASIC DISPLAY PARAMETERS (b GROUP)

The b group parameters are read-only and display real-time operating data:

b001 Output Freq ? Current output frequency to the motor in Hz.
b002 Commanded Freq ? The frequency command value in Hz.
b003 Output Current ? Motor output current in Amps.
b004 Output Voltage ? Motor output voltage in Volts.
b005 DC Bus Voltage ? DC bus voltage in Volts (nominal 650V DC for 480V input).
b006 Drive Status ? Current drive state (Ready, Run, Fault, etc.).
b007 Fault 1 Code ? Most recent fault code.
b008 Fault 2 Code ? Second most recent fault code.
b009 Fault 3 Code ? Third most recent fault code.
b010 Process Fract ? Process variable fractional display.
b011 Process Units ? Process variable in engineering units.
b012 Control Source ? Active control source (HIM, Network, Terminal, etc.).
b013 Control In Status ? Digital input status display.
b014 Dig In Status ? Terminal block digital input status.
b015 Dig Out Status ? Terminal block digital output status.
b016 Speed Feedback ? Actual motor speed feedback in RPM.
b017 Elapsed Run Time ? Total run time in hours.
b018 Power Saved ? Estimated power savings in MWh.
b019 Accum MWh ? Total accumulated energy in MWh.
b020 Accum Cost Sav ? Estimated energy cost savings.

2. INSTALLATION

2.1 ENVIRONMENTAL REQUIREMENTS

Ambient temperature: 0°C to 50°C (derate to 90% above 40°C)
Humidity: 5% to 95% non-condensing
Altitude: up to 1000 m without derating, 1% per 100 m above 1000 m
Enclosure: IP20 (NEMA 1) standard, IP54 option available

2.2 POWER WIRING

Use copper wire rated for 75°C minimum. Tighten all power terminals to the torque specified in Table 2.1. Route power wiring and control wiring in separate conduits, separated by at least 150 mm. If crossing is unavoidable, cross at a 90° angle.

2.3 CONTROL WIRING

Control terminals accept 0.5 mm² to 2.5 mm² wire. Use shielded twisted pair cable for analog and communication signals. Connect the shield at the drive end only. Ground the shield via the shield clamp inside the drive enclosure.

2.4 FUSE AND CIRCUIT BREAKER RATINGS

Catalog 25B-D1P4N104 (0.4 kW): 6A class CC fuse
Catalog 25B-D6P0N104 (3.0 kW): 20A class CC fuse
Catalog 25B-D019N104 (9.0 kW): 50A class J fuse
Catalog 25B-D480N104 (75 kW): 200A class J fuse

3. PROGRAMMING

3.1 ADVANCED PROGRAM GROUP (A GROUP)

A410 Preset Freq 0 ? First preset frequency reference.
A411 Preset Freq 1 ? Second preset frequency reference.
A444 Accel Time 3 ? Third acceleration ramp time in seconds.
A445 Decel Time 3 ? Third deceleration ramp time in seconds.
A479 PID 2 Invert Err ? Invert the PID 2 error signal.
A481 Process Disp Lo ? Low value for process display scaling.
A519 PM HIFI NS Cur(1) ? Permanent magnet motor nominal current.
A520 PM Bus Reg Kd(1) ? PM motor bus regulator derivative gain.
A554 Drv Ambient Sel ? Drive ambient temperature sensor selection.
A555 Reset Meters ? Reset accumulated energy and run time meters.

3.2 COMMUNICATIONS GROUP (C GROUP)

C143 Comm Flt Actn ? Action on communication fault (0=Stop, 1=Hold, 2=Alarm).
C156 EN Data In 4 ? EtherNet/IP data input word 4.
C157 Opt Data Out 3 ? Option module data output word 3.
C158 EN Rcv Data 1 ? Ethernet received data word 1.

4. FAULT AND DIAGNOSTIC INFORMATION

4.1 FAULT CODE DESCRIPTIONS

F004 ? UnderVoltage (DC bus voltage below 400V DC)
Meaning: The DC bus voltage has fallen below the undervoltage threshold of 400V DC. This is typically caused by a loss of input AC power, a blown input fuse, or a severely sagging AC line voltage.

F005 ? OverVoltage (DC bus voltage above 815V DC)
Meaning: The DC bus voltage has exceeded the overvoltage threshold of 815V DC. This is typically caused by excessive regenerative energy from the motor during deceleration, or a transient overvoltage on the AC input.

F006 ? Motor Stalled
Meaning: The motor rotor has stalled or is rotating at less than 25% of the commanded speed. The stall condition must persist for the stall time set in parameter P-36 (default 3 seconds) before the fault is triggered.

F008 ? Motor Overload
Meaning: The motor current has exceeded the overload current rating set in parameter P-35 (default 150% of motor FLA) for longer than the allowable time. The drive uses an I^2t thermal model to calculate motor heating.

F009 ? Drive Overload
Meaning: The drive output current has exceeded 110% of the drive's rated output current. The drive's internal I^2t overload protection has tripped.

F012 ? Ground Fault
Meaning: A ground fault has been detected on the drive output. The drive senses current imbalance between the three output phases exceeding 50% of the rated current. Check motor wiring insulation.

F013 ? Hardware OverCurrent
Meaning: Instantaneous peak current exceeded 200% of the drive rating. This is a hardware-level fault, not a software overload. Check for a shorted motor or damaged output IGBT.

F071 ? PI Feedback Loss
Meaning: The PID feedback signal has fallen below the loss threshold set in parameter A-469 for longer than the loss time set in A-470.

F080 ? Safe Torque Off
Meaning: The Safe Torque Off (STO) circuit has been activated. Check the STO input terminals and the safety system.

4.2 FAULT HISTORY PARAMETERS

The drive records the last 10 faults with time and operating data:
F604 Fault 4 Code ? Fourth fault code.
F605 Fault 5 Code ? Fifth fault code.
F606 Fault 6 Code ? Sixth fault code.
F607 Fault 7 Code ? Seventh fault code.
F608 Fault 8 Code ? Eighth fault code.
F609 Fault 9 Code ? Ninth fault code.
F610 Fault 10 Code ? Tenth fault code.

Each fault record also stores:
F627 Fault 1 Time-min ? Fault 1 occurrence time in minutes.
F628 Fault 2 Time-min ? Fault 2 occurrence time in minutes.
F629 Fault 3 Time-min ? Fault 3 occurrence time in minutes.

F651 Fault 1 BusVolts ? DC bus voltage at the time of Fault 1.
F652 Fault 2 BusVolts ? DC bus voltage at the time of Fault 2.
F653 Fault 3 BusVolts ? DC bus voltage at the time of Fault 3.

5. TROUBLESHOOTING

5.1 DRIVE DOES NOT START

Check: AC power present (LED D1 green), control power present, enable input closed, no active faults, speed reference above 0 Hz.

5.2 MOTOR RUNS ERRATICALLY

Check: motor data programmed correctly (P-01 through P-06), autotune completed (P-40), input voltage stable, ground faults absent.

5.3 DRIVE OVERHEATING

Check: cooling fan operation (D1 on the control board), ambient temperature, air filter, drive loading not exceeding rated current.

6. MAINTENANCE

6.1 COOLING FAN REPLACEMENT

The internal cooling fan (P/N PF-525-FAN) should be replaced every 40,000 hours of operation. The fan is located behind the front cover, secured by two M4 screws. Disconnect the fan connector from the control board (J14) before removing.

6.2 CAPACITOR FORMATION

If the drive has been stored for more than 12 months without power, the DC bus capacitors must be reformed before applying full power. Apply 50% of the rated voltage for 30 minutes, then 75% for 30 minutes, before applying full power.